Eugene Ruiz Classes
CRC Cards & Class Details
Ver.1.4

CRC Cards

Class Name	Responsibilities	Collaborators
Flight	• Store flight data (number, origin, destination, terminals, times, status, aircraft) • Provide access to all attributes via getters/setters • Return available seats • Update status • Return formatted flight details	**Airport, Terminal, Schedule, Aircraft, Booking**
Schedule	• Store schedule data (ID, departure/arrival times) • Provide access to all attributes • Manage list of flights (add/remove) • Return schedule info	**Flight**
Booking	• Store booking data (ID, date, status, tickets) • Provide access to all attributes • Add/cancel tickets • Return booking status	**Ticket, Payment, Passenger**
Ticket	• Store ticket data (ID, flight, seat, status, passenger) • Provide access to all attributes • Assign/cancel seat/ticket • Return formatted ticket details • Provide passenger reference (Group 2 support)	**Flight, Seat, Passenger, Booking, TicketManager**
Payment	• Store payment data (ID, booking, amount, date, method, status) • Provide access to all attributes • Process payment • Update status and check completion	**Booking**
BookingManager	• Maintain collection of all bookings • Create, remove, find bookings • Get bookings for a specific passenger	**Booking, Passenger**
TicketManager	• Maintain collection of all tickets • Create, remove, find tickets • Provide getAllTickets() and getTicketsForPassenger() (Group 2 support)	**Ticket, Passenger**
Airline	• Hold references to all managers (light coordinator only) • Provide access to managers • Return airline name	**AirportManager, StaffManager, BookingManager, TicketManager**

Class Details

Flight

Instance Variables:
flightNumber: String Unique identifier for the flight (e.g., "AA123")
origin: Airport Departure airport for this flight
destination: Airport Arrival airport for this flight
departureTerminal: Terminal Terminal at the origin airport
arrivalTerminal: Terminal Terminal at the destination airport
departureTime: Schedule Schedule object containing departure timing
arrivalTime: Schedule Schedule object containing arrival timing
status: String Current flight status (e.g., "ON_TIME", "DELAYED", "CANCELLED")
aircraft: Aircraft Aircraft assigned to this flight

Methods:
Flight(String flightNumber, Airport origin, Airport destination, Terminal departureTerminal, Terminal arrivalTerminal, Schedule departureTime, Schedule arrivalTime, String status, Aircraft aircraft) • Full constructor that initializes all flight data • Ensures the object is always in a valid state
getFlightNumber(): String • Returns the unique flight number
setFlightNumber(String flightNumber): void • Updates the flight number
getOrigin(): Airport • Returns the departure airport
setOrigin(Airport origin): void

• Updates the departure airport
getDestination(): Airport • Returns the arrival airport
setDestination(Airport destination): void • Updates the arrival airport
getDepartureTerminal(): Terminal • Returns the departure terminal
setDepartureTerminal(Terminal departureTerminal): void • Updates the departure terminal
getArrivalTerminal(): Terminal • Returns the arrival terminal
setArrivalTerminal(Terminal arrivalTerminal): void • Updates the arrival terminal
getDepartureTime(): Schedule • Returns the departure schedule
setDepartureTime(Schedule departureTime): void • Updates the departure schedule
getArrivalTime(): Schedule • Returns the arrival schedule
setArrivalTime(Schedule arrivalTime): void • Updates the arrival schedule
getStatus(): String • Returns the current flight status
setStatus(String status): void • Updates the flight status
getAircraft(): Aircraft • Returns the assigned aircraft
setAircraft(Aircraft aircraft): void • Updates the assigned aircraft
getAvailableSeats(): int • Returns the number of available seats on this flight
updateStatus(String newStatus): void • Updates the flight status with a new value
getFlightDetails(): String • Returns a formatted string with key flight information

Schedule

Instance Variables:
scheduleID: String Unique identifier for the schedule
departureTime: String Departure time for flights on this schedule
arrivalTime: String Arrival time for flights on this schedule
scheduleID: String Unique identifier for the schedule

Methods:
Schedule(String scheduleID, String departureTime, String arrivalTime) • Constructor that initializes schedule timing data
getScheduleID(): String • Returns the unique schedule identifier
setScheduleID(String scheduleID): void • Updates the schedule identifier
getDepartureTime(): String • Returns the departure time
setDepartureTime(String departureTime): void • Updates the departure time
getArrivalTime(): String • Returns the arrival time
setArrivalTime(String arrivalTime): void • Updates the arrival time
addFlight(Flight f): void • Adds a flight to this schedule

Booking

Instance Variables:
bookingID: String Unique identifier for the booking
bookingDate: Date Date the booking was created
status: String Current status of the booking (e.g., "CONFIRMED")
tickets: ArrayList<Ticket> List of tickets included in this booking

Methods:
Booking(String bookingID, Date bookingDate, String status) • Constructor that initializes booking data
getBookingID(): String • Returns the unique booking identifier
setBookingID(String bookingID): void • Updates the booking identifier
getBookingDate(): Date • Returns the booking creation date
setBookingDate(Date bookingDate): void • Updates the booking date
getStatus(): String • Returns the current booking status
setStatus(String status): void • Updates the booking status
getTickets(): ArrayList<Ticket> • Returns the list of tickets in this booking
addTicket(Ticket t): void • Adds a ticket to this booking
cancelBooking(): void • Cancels the entire booking
getBookingStatus(): String • Returns the current status of the booking

Booking(String bookingID, Date bookingDate, String status) • Constructor that initializes booking data
getBookingID(): String • Returns the unique booking identifier
setBookingID(String bookingID): void • Updates the booking identifier
getBookingDate(): Date • Returns the booking creation date
setBookingDate(Date bookingDate): void • Updates the booking date
getStatus(): String • Returns the current booking status

Ticket

Instance Variables:
ticketID: String Unique identifier for the ticket
flight: Flight Flight this ticket is booked for
seat: Seat Seat reserved on the flight
status: String Current status of the ticket
passenger: Passenger Passenger who owns this ticket [**supports Passenger class's method for viewTickets()**]

Methods:
Ticket(String ticketID, Flight flight, Seat seat, String status, Passenger passenger) • Full constructor that initializes all ticket data
getTicketID(): String • Returns the unique ticket identifier
setTicketID(String ticketID): void • Updates the ticket identifier
getFlight(): Flight • Returns the associated flight
setFlight(Flight flight): void • Updates the associated flight
getSeat(): Seat • Returns the reserved seat
setSeat(Seat seat): void • Updates the reserved seat
getStatus(): String • Returns the current ticket status
setStatus(String status): void • Updates the ticket status
getPassenger(): Passenger

• Returns the passenger who owns this ticket
setPassenger(Passenger passenger): void • Updates the owning passenger
assignSeat(Seat s): void • Assigns (or re-assigns) a seat to this ticket
cancelTicket(): void • Cancels the ticket by updating its status
getTicketDetails(): String • Returns formatted ticket details for display

Payment

Instance Variables:
paymentId: String Unique identifier for the payment
booking: Booking Booking this payment is associated with
amount: double Payment amount
paymentDate: Date Date the payment was made
method: String Payment method (e.g., "credit_card")
status: String Current payment status

Methods:
Payment(Booking booking, double amount, String method) • Constructor that initializes payment data
getPaymentId(): String • Returns the unique payment identifier
setPaymentId(String paymentId): void • Updates the payment identifier
getBooking(): Booking • Returns the associated booking
setBooking(Booking booking): void • Updates the associated booking
getAmount(): double • Returns the payment amount
setAmount(double amount): void • Updates the payment amount
getPaymentDate(): Date • Returns the payment date

BookingManager

Instance Variables:
bookings: ArrayList Collection of all Booking objects in the system

Methods:
BookingManager() • Constructor that initializes the booking collection
createBooking(Passenger p): Booking • Creates a new booking for the given passenger
removeBooking(String bookingID): void • Removes a booking by its ID
findBooking(String bookingID): Booking • Finds and returns a booking by its ID
getBookingsForPassenger(Passenger p): List<Booking> • Returns all bookings belonging to a specific passenger

TicketManager

Instance Variables:
tickets: ArrayList<Ticket> Collection of all Ticket objects in the system

Methods:
TicketManager() • Constructor that initializes the ticket collection
createTicket(Passenger p, Flight f): Ticket • Creates a new ticket and adds it to the collection
removeTicket(Ticket t): void • Removes a ticket from the collection
findTicket(String ticketID): Ticket • Finds and returns a ticket by its ID

Airline

Instance Variables:
airlineName: String Name of the airline
airportManager: AirportManager Reference to the airport management component
staffManager: StaffManager Reference to the staff management component
bookingManager: BookingManager Reference to the booking management component
ticketManager: TicketManager Reference to the ticket management component

Methods:
Airline(String airlineName) • Constructor that initializes the airline with its name
getAirportManager(): AirportManager • Returns the airport manager reference
getStaffManager(): StaffManager • Returns the staff manager reference
getBookingManager(): BookingManager • Returns the booking manager reference
getTicketManager(): TicketManager • Returns the ticket manager reference
getAirlineName(): String • Returns the airline name